

Seongmoon Jeong

PHD CANDIDATE

IRIS Lab., Sungkyunkwan University, Suwon, Republic of Korea

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research centers on **image compression** within computer vision, spanning both low-level optimization and its impact on high-level tasks such as classification, object detection, and segmentation. While conventional compression is designed for human perception, I investigate how visual information can be encoded to better serve **machine perception**. In particular, I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have been working on high-level vision directly in the **RAW domain**, prior to the image signal processing (ISP) pipeline, to better preserve information relevant to machine perception. My long-term goal is to enable **communication-efficient intelligent systems** for real-world applications, including **autonomous systems and edge-based machine vision**.

Education

Ph.D. in Artificial Intelligence

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Current

- Got a Outstanding Scholarship which is given to promising students in AI Dept.

B.S. in Electronic and Electrical Engineering

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2013 - Feb. 2020

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Current

- General Codec Adapter for Machines
- Learned Image Compression Adapter for Machines
- Efficient JPEG Modification for Machines

Project Experience

Video Coding for Machines

LEAD RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2023 - 2024

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images

LEAD RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2022

Selected Publications

Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

1ST AUTHOR

IEEE AVSS

Sep. 2026

UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

1ST AUTHOR

IEEE AVSS

Sep. 2026

Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

2ND AUTHOR

IEEE CVPR

Mar. 2025

Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

2ND AUTHOR

IEEE AICAS

Feb. 2024

Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

1ST AUTHOR

IEEE ICME Workshop

Apr. 2023

An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

2ND AUTHOR

Pattern Recognition Letters

Nov. 2022

Standardization Activities

[VCM] Neural network based post-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71305

Geneva

Jan. 2025

[VCM] Neural network based pre-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71304

Geneva

Jan. 2025

[VCM] Learned joint filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69990

Kemer

Nov. 2024

[VCM] Learned pre-filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69085

Sapporo

Jan. 2024

Honors & Awards

2024 **3rd Place**, DAC System Design Contest - GPU Track

San Francisco, U.S.A

2020 **2nd Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

2019 **Honorable Mention**, Big Data Center Open Innovation Challenge, Seongnam-si

Seongnam, Korea

2019 **Outstanding Work Award**, Capstone Design Project, SKKU

Suwon, Korea

2018 **Dean's List Award**, Fall semester, SKKU

Suwon, Korea

2017 **Dean's List Award**, Fall semester, SKKU

Suwon, Korea

Full Publications

INTERNATIONAL CONFERENCE

Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

1ST AUTHOR

IEEE AVSS

Sep. 2026

UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

1ST AUTHOR

IEEE AVSS

Sep. 2026

Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

2ND AUTHOR

IEEE CVPR

Mar. 2025

Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

2ND AUTHOR

IEEE AICAS

Feb. 2024

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays

4TH AUTHOR

IEEE ICCAD

Jul. 2023

INTERNATIONAL CONFERENCE (CONTINUED)

Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

1ST AUTHOR

IEEE ICME Workshop

Apr. 2023

INTERNATIONAL JOURNAL

A Codec-Transferable Adapter for Machine-Oriented Image Compression Across Multiple Vision Tasks

1ST AUTHOR

Preprint (Under review for TCSVT)

2025

KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping

4TH AUTHOR

IEEE TCAS-I

Feb. 2024

An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

2ND AUTHOR

Pattern Recognition Letters

Nov. 2022

Student Mentoring

Hangyul Choi

M.S. STDUENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS

2023 - 2024

- Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Chanung Park

M.S. STDUENT AT SUNGKYUNKWAN UNIVERSITY

2024

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, Tensorflow, JAX, TensorRT
Development Frameworks	Docker, Conda, Uv

Reference

Jong Hwan Ko

ASSOCIATE PROFESSOR

Ph.D. advisor

Dept. of ECE, SKKU, Suwon, Korea

- <https://iris.skku.edu/>
- jhko@skku.edu